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Recap – what are embedded systems

Peripherals
- ADC
- Timers
- GPIOs
- Interrupts



User interface
(HMI/UX)

Inputs
- Monitoring the environment
- User inputs

Processing
- Hardware
- Firmware

Outputs
- Interact with the environment or user

Connectivity

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Difference between MCU & MPU

Feature	Microcontroller (MCU)	Microprocessor (MPU)
Integration	All-in-one chip: includes CPU, memory, I/O, clock, and power management	Only CPU on chip; needs external memory, I/O controllers, clock, and power circuitry
Processing Power	Moderate — ideal for control-oriented tasks	High — suitable for heavy, complex data processing
Power Consumption	Low power, optimized for energy efficiency and battery operation	High power usage; often needs cooling/thermal management
Cost & Complexity	Low cost, compact, and simpler design	Higher cost and complexity due to need for external components
Security & Reliability	Integrated lightweight security (good for embedded)	Advanced scalable security for high-performance environments
Real-Time Processing	Excellent for real-time tasks using RTOS or bare-metal	Supports full OS (Linux, Windows); real-time via hypervisors
Applications	Appliances, smart devices, sensors, automotive systems	Laptop/desktop computers, tablets, servers, multimedia systems, edge computing

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Summary

Please click the table to enlarge it



Criteria	What to Consider	Impact
Application	Task complexity, environment	Determines MCU performance and features
Power Consumption	Battery vs mains power, sleep modes	Affects battery life and efficiency
Architecture & Bit Size	8-bit, 16-bit, 32-bit; Harvard vs Von Neumann	Influences speed, power, and complexity
Hardware Interfaces	Required communication and I/O peripherals	Ensures compatibility with other devices
Memory	Flash and RAM size	Supports application code and data
Cost & Availability	Price, supply chain stability	Affects project budget and production
Development Support	SDKs, libraries, debugging tools, community support	Reduces development time and effort
Special Features	Security, timers, real-time clock	Adds functionality and future-proofing

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Renesas RA6E2 microcontroller

RA6E2 200MHz 32 Bit Arm® Cortex®- M33 Core NVIC | SWD | ETB

Communication

- USB2.0 FS Device x 1*
- CAN FD x1
- I3C x1
- SCI x2
- SPI x2
- HDMI CEC x1
- SSI x1
- QSPI x1

* Excluding 32-pin package

Type	Protocol	Use Case
Serial (1-to-1)	UART (Universal Asynchronous Receiver Transmitter)	Communication between two devices, like a microcontroller and GPS module
Synchronous Serial (1-to-many)	SPI (Serial Peripheral Interface)	Fast communication with multiple sensors or displays
	I^2C(Inter-Integrated Circuit), I^3C	Communicating with multiple devices over just two wires, like EEPROMs, RTCs, etc.
Wireless	8-bit, 16-bit, 32-bit; Harvard vs Von Neumann	8-bit, 16-bit, 32-bit; Harvard vs Von Neumann
Network	Required communication and I/O peripherals	Required communication and I/O peripherals

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- From Simple I/O to Complex Systems
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- The Designer's Dilemma: A Balancing ACt
- An Industrial perspective: Selecting the Right MCU
- The MCU as a Brain: Interfacing with the world
- GPIO: The Hands and feet
- ADC: The senses of Sight, Hearing and Touch
- Timers: The internal Heartbeat & Modes
- Timer Flavors: The Workhorse and the Marathon Runner
- The System's Guardian Watchdog Timer (WDT)
- Standard Interfaces : The Language Centers
- From Whispers to Networks: Mastering Communication Protocols
- End of Module

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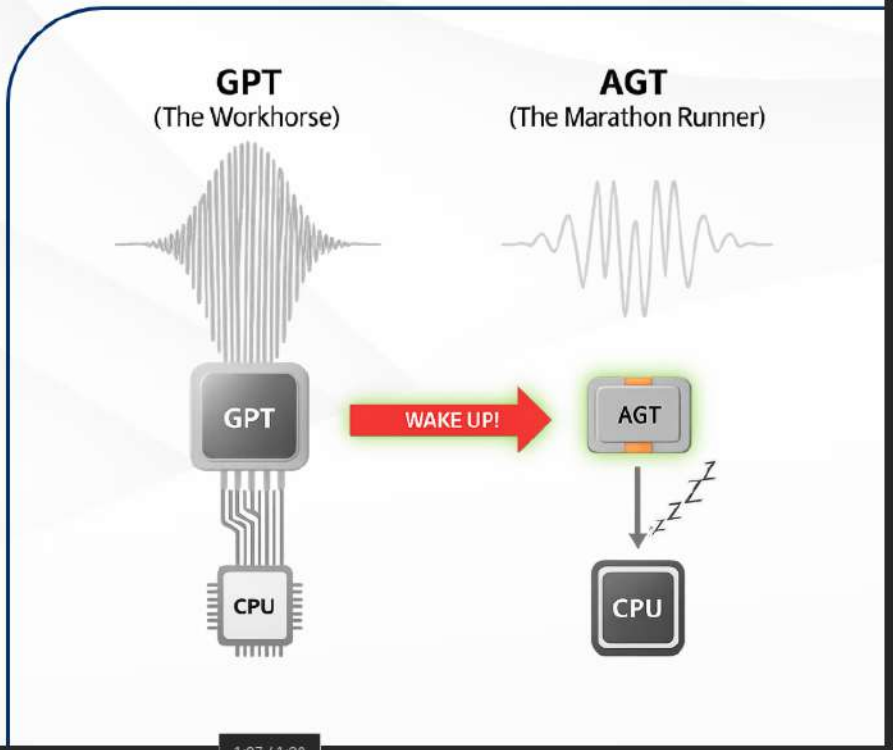
Timer Flavors: The Workhorse and the Marathon Runner

General Purpose Timer (GPT): The Workhorse

- A versatile 16-bit or 32-bit timer synchronized with the main high-speed peripheral clock.
- Ideal for high-resolution timing, fast PWM, and input capture.
- Many MCUs, like the Renesas RA series, feature multiple GPT channels.

Asynchronous General Purpose Timer (AGT): The Marathon Runner

- A specialized low-power timer found in Renesas RA MCUs.
- Can run from a separate, low-frequency clock (e.g., 32kHz), independent of the main CPU clock.
- **Key Advantage:** The AGT can remain active to wake the system from deep sleep modes, enabling ultra-low-power applications.



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The System's Guardian Watchdog Timer (WDT)

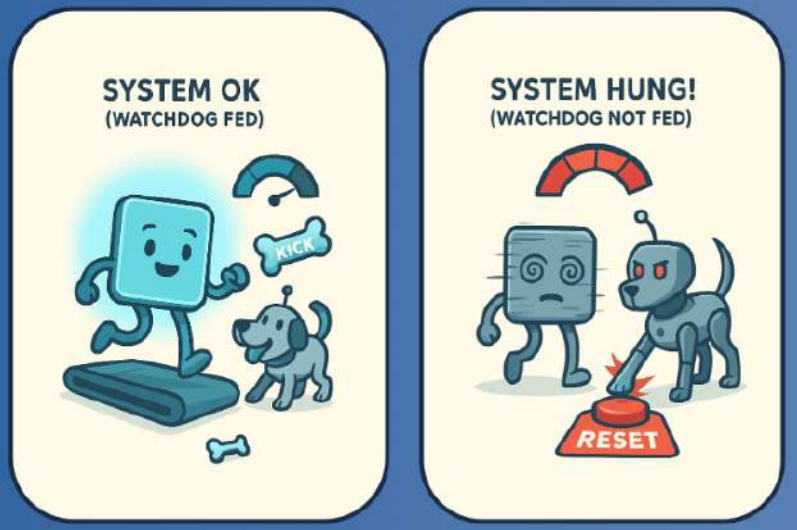
Function: A safety mechanism, not a general timer. It's a hardware counter that protects a system from software faults (e.g., an infinite loop).

How it Works

- The software must periodically "feed" or "kick" the watchdog timer to reset its counter.
- If the software hangs and fails to feed the dog, the timer will expire (overflow).
- Upon expiration, the watchdog forces a hardware reset of the entire MCU, returning it to a known good state.

Why it's Critical

- Essential for reliable, unattended systems.
- It's the last line of defense against a system becoming completely unresponsive.



PREV NEXT

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Standard Interfaces : The Language Centers

Function: Provide standardized protocols (rules of grammar) for the MCU to "speak" with other complex chips and systems.

Why not just use GPIO? Communicating complex data using only GPIOs ("bit-banging") is slow, inefficient, and burdens the CPU. Hardware interfaces handle the protocol automatically.

UART

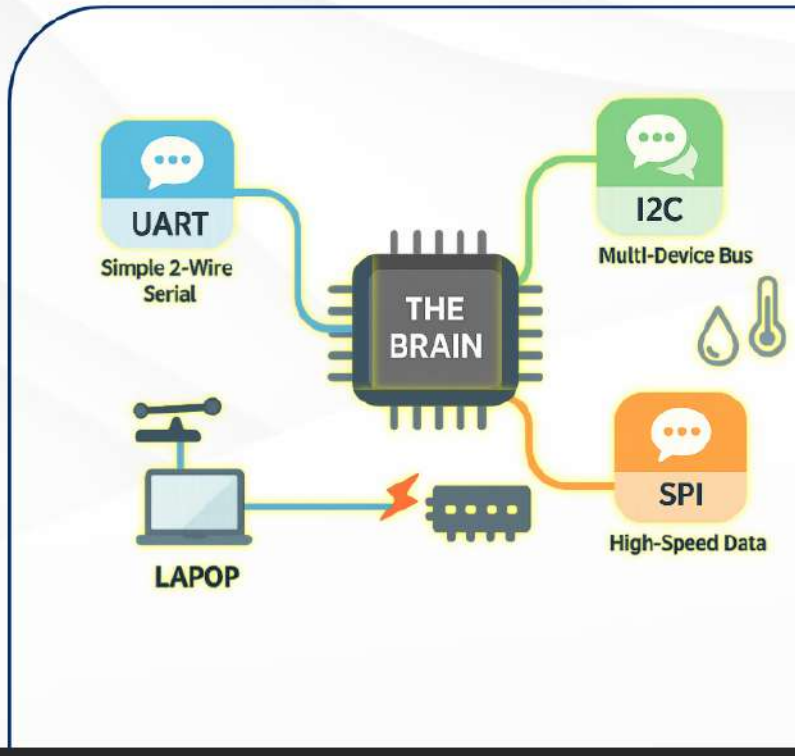
Asynchronous serial. Like a simple, two-wire telegraph. Great for PC communication, GPS modules.

I2C

A two-wire bus for connecting one "master" MCU to multiple "slave" sensors on the same wires.

SPI

A fast, synchronous bus for high-bandwidth devices like displays and flash memory.



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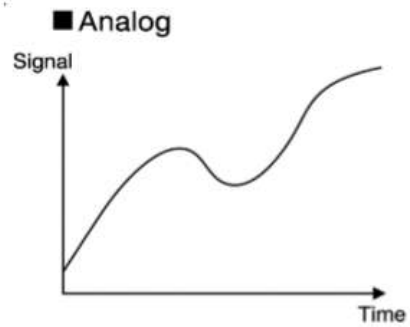
The Two Worlds: Analog vs. Digital Signals

Analog Signals

- **Nature:** Continuous in both time and amplitude.
- **Representation:** Represents physical phenomena directly (e.g., voltage, pressure, sound).
- **Characteristics**: Infinite resolution within a given range, but highly susceptible to noise and degradation during transmission and processing.
- **Waveform**: Sine wave.

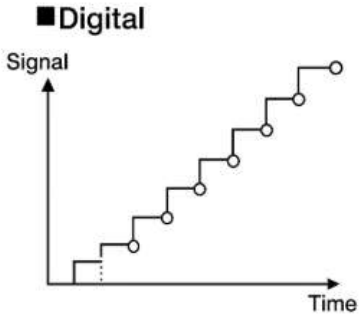
Digital Signals

- **Nature**: Discrete in time (sampled) and amplitude (quantized).
- **Representation**: Data encoded as a sequence of discrete values, typically binary (0s and 1s).
- **Characteristics**: Finite resolution, but robust against noise. Ideal for computation, storage, and error correction.
- **Waveform**: Square wave.



An analog circuit operates on continuous signals.

Physical quantities – such as voltage, current, and time - change continuously.



A digital circuit operates on discrete (discontinuous) signals.

Physical qualities take discrete values only.